

0.1 General Definition

Previously, our discussion revolved around the finite field $\mathbb{F}_p$ for a prime p . However, many protocols need more than just a prime field. For example, elliptic curve pairings and certain STARK constructions require extending $\mathbb{F}_p$ to, in a sense, the analogous of complex numbers.

From school and, possibly, university, you might remember how complex numbers $\mathbb{C}$ are constructed. You take two real numbers, say, $x, y \in \mathbb{R}$, introduce a new symbol i satisfying $i^2 = -1$, and define the complex number as $z = x + iy$. In certain cases, one might encounter a bit more rigorous and abstract definition of complex numbers as the set of pairs $(x, y) \in \mathbb{R}^2$ where addition is naturally defined as $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$, and the multiplication is:

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1)^1.$$

In spite of what interpretation you have seen, the complex number is just a tuple of two real numbers that satisfy a bit different rules of multiplication (since addition is typically defined in the same way). What is even more important to us, is that $\mathbb{C}$ is our first example of the so-called **field extension** of $\mathbb{R}$.

Formally, definition of the field extension is straightforward:

Definition 0.1. Let $\mathbb{F}$ be a field and $\mathbb{K}$ be another field. We say that $\mathbb{K}$ is an **extension** of $\mathbb{F}$ if $\mathbb{F} \subset \mathbb{K}$ and we denote it as $\mathbb{K}/\mathbb{F}$.

Despite just a simplicity of the definition, the field extensions are a very powerful tool in mathematics. But first, let us consider a few non-trivial examples of field extensions.

Example 0.1. Denote by $\mathbb{Q}(\sqrt{2}) = \{x + y\sqrt{2} : x, y \in \mathbb{Q}\}$. This is a field extension of $\mathbb{Q}$. It is obvious that $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$, but why is $\mathbb{Q}(\sqrt{2})$ a field? Addition and multiplication operations are obviously closed:

$$\begin{aligned} (x_1 + y_1\sqrt{2}) + (x_2 + y_2\sqrt{2}) &= (x_1 + x_2) + (y_1 + y_2)\sqrt{2}, \\ (x_1 + y_1\sqrt{2}) \cdot (x_2 + y_2\sqrt{2}) &= (x_1x_2 + 2y_1y_2) + (x_1y_2 + x_2y_1)\sqrt{2}. \end{aligned}$$

But what about the inverse element? Well, here is the trick:

$$\begin{aligned} \frac{1}{x + y\sqrt{2}} &= \frac{x - y\sqrt{2}}{(x + y\sqrt{2})(x - y\sqrt{2})} = \frac{x - y\sqrt{2}}{x^2 - 2y^2} = \\ &= \frac{x}{x^2 - 2y^2} - \frac{y}{x^2 - 2y^2}\sqrt{2} \in \mathbb{Q}(\sqrt{2}). \end{aligned}$$

Example 0.2. Consider $\mathbb{Q}(\sqrt{2}, i) = \{a + bi : a, b \in \mathbb{Q}(\sqrt{2})\}$ where $i^2 = -1$. This is a field extension of $\mathbb{Q}(\sqrt{2})$ and, consequently, of $\mathbb{Q}$. The representation of the element is:

¹Notice that $(x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + iy_2x_1 + iy_1x_2 + i^2y_1y_2 = (x_1x_2 - y_1y_2) + (x_1y_2 + x_2y_1)i$.

$$(a + b\sqrt{2}) + (c + d\sqrt{2})i = a + b\sqrt{2} + ci + d\sqrt{2}i$$

Showing that this is a field is a bit more tedious, but still straightforward. Suppose we take $\alpha + \beta i \in \mathbb{Q}(\sqrt{2}, i)$ with $\alpha, \beta \in \mathbb{Q}(\sqrt{2})$. Then:

$$\frac{1}{\alpha + \beta i} = \frac{\alpha - \beta i}{\alpha^2 + \beta^2} = \frac{\alpha}{\alpha^2 + \beta^2} - \frac{\beta}{\alpha^2 + \beta^2}i$$

Since $\mathbb{Q}(\sqrt{2})$ is a field, both $\frac{\alpha}{\alpha^2 + \beta^2}$ and $\frac{\beta}{\alpha^2 + \beta^2}$ are in $\mathbb{Q}(\sqrt{2})$, and, consequently, $\mathbb{Q}(\sqrt{2}, i)$ is a field as well.

Remark. Notice that basically, $\mathbb{Q}(\sqrt{2}, i)$ is just a linear combination of $\{1, \sqrt{2}, i, \sqrt{2}i\}$. This has a very important implication: $\mathbb{Q}(\sqrt{2}, i)$ is a four-dimensional vector space over $\mathbb{Q}$, where elements $\{1, \sqrt{2}, i, \sqrt{2}i\}$ naturally form **basis**. We are not going to use it implicitly, but this observation might make further discussion a bit more intuitive.

Remark. One might have defined $\mathbb{Q}(\sqrt{2}, i) = \{x + \sqrt{2}y : x, y \in \mathbb{Q}(i)\}$ instead. Indeed, $\mathbb{Q}(\sqrt{2})(i) = \mathbb{Q}(i)(\sqrt{2}) = \mathbb{Q}(\sqrt{2}, i)$.

0.2 Polynomial Quotient Ring

Now, we present a more general way to construct field extensions. Notice that when constructing $\mathbb{C}$, we used the magical element i that satisfies $i^2 = -1$. But here is another way how to think of it.

Consider the set of polynomials $\mathbb{R}[x]$, then I pick $p(x) := x^2 + 1 \in \mathbb{R}[x]$ and ask you to find roots of $p(x)$. Of course, you would claim “hey, this equation has no solutions over $\mathbb{R}$ ” and that is totally true. That is why mathematicians introduced a new element i that we formally called the root of $x^2 + 1$. Note however, that i is not a number in the traditional sense, but rather a fictional symbol that we artificially introduced to satisfy the equation.

Now, could we have picked another polynomial, say, $q(x) = x^2 + 4$? Sure! As long as its roots cannot be found in $\mathbb{R}$, we are good to go.

Example 0.3. Suppose β is the root of $q(x) := x^2 + 4$. Then we could have defined complex numbers as a set of $x + y\beta$ for $x, y \in \mathbb{R}$. In this case, multiplication, for example, would be defined a bit differently than in the case of $\mathbb{C}$:

$$(x_1 + y_1\beta) \cdot (x_2 + y_2\beta) = (x_1x_2 - 4y_1y_2) + (x_1y_2 + x_2y_1)\beta.$$

We shifted to the polynomial consideration for a reason: now, instead of considering the complex number $\mathbb{C}$ as “some” tuple of real numbers (c_0, c_1) , now let us view it as a polynomial² $c_0 + c_1X$ modulo polynomial $X^2 + 1$.

²Here, we use X to represent the polynomial variable to avoid confusion with the notation $x + yi$.

Example 0.4. Indeed, take, for example, $p_1(X) := 1 + 2X$ and $p_2(X) := 2 + 3X$. Addition is performed as we are used to:

$$p_1 + p_2 = (1 + 2X) + (2 + 3X) = 3 + 5X,$$

but multiplication is a bit different:

$$p_1 p_2 = (1 + 2X) \cdot (2 + 3X) = 2 + 3X + 4X + 6X^2 = 6X^2 + 7X + 2.$$

Well, and what next? Recall that we are doing arithmetic modulo $X^2 + 1$ and for that reason, we divide the polynomial by $X^2 + 1$:

$$6X^2 + 7X + 2 = 6(X^2 + 1) + 7X - 4 \implies (6X^2 + 7X + 2) \bmod (X^2 + 1) = 7X - 4,$$

meaning that $p_1 p_2 = 7X - 4$. Oh wow, hold on! Let us come back to our regular complex number representation and multiply $(1 + 2i)(2 + 3i)$. We get $2 + 3i + 4i + 6i^2 = -4 + 7i$. That is exactly the same result if we change X to i above! In fact, what we have observed is the fact that our polynomial quotient ring $\mathbb{R}[X]/(X^2 + 1)$ is isomorphic to $\mathbb{C}$.

So, let us generalize this observation to any field $\mathbb{F}$ and any irreducible polynomial $\mu(x) \in \mathbb{F}[x]$.

Theorem 0.2. Let $\mathbb{F}$ be a field and $\mu(x)$ — irreducible polynomial over $\mathbb{F}$ (sometimes called a **reduction polynomial**). Consider a set of polynomials over $\mathbb{F}[x]$ modulo $\mu(x)$, formally denoted as $\mathbb{F}[x]/(\mu(x))$. Then, $\mathbb{F}[x]/(\mu(x))$ is a field.

Example 0.5. As we considered above, let $\mathbb{F} = \mathbb{R}$, $\mu(x) = x^2 + 1$, then $\mathbb{R}[X]/(X^2 + 1)$ (a set of polynomials modulo $X^2 + 1$) is a field.

Example 0.6. Suppose $\mathbb{F} = \mathbb{Q}$ and $\mu(x) := x^2 - 2$. Then, $\mathbb{Q}[X]/(X^2 - 2)$ is a field isomorphic to $\mathbb{Q}(\sqrt{2})$, considered above.

Example 0.7. Suppose $\mathbb{F} = \mathbb{Q}$ and $\mu(x) := (x^2 + 1)(x^2 - 2) = x^4 - x^2 - 2$. Then, $\mathbb{Q}[X]/(x^4 - x^2 - 2)$ is a field isomorphic to $\mathbb{Q}(\sqrt{2}, i)$.

Remark. Although we have not defined the isomorphism between two rings/fields, it is defined similarly to group isomorphism. Suppose we have fields $(\mathbb{F}, +, \times)$ and $(\mathbb{K}, \oplus, \otimes)$. Bijective function $\phi : \mathbb{F} \rightarrow \mathbb{K}$ is called an isomorphism if it preserves additive and multiplicative structures, that is for all $a, b \in \mathbb{F}$:

$$\phi(a + b) = \phi(a) \oplus \phi(b),$$

$$\phi(a \times b) = \phi(a) \otimes \phi(b).$$

This theorem (aka definition) corresponds to viewing complex numbers as

a polynomial quotient ring $\mathbb{R}[X]/(X^2 + 1)$. But, we can give a theorem (aka definition) for our classical representation via magical root i of $x^2 + 1$.

Theorem 0.3. Let $\mathbb{F}$ be a field and $\mu \in \mathbb{F}[X]$ is an irreducible polynomial of degree n and let $\mathbb{K} := \mathbb{F}[X]/(\mu(X))$. Let $\theta \in \mathbb{K}$ be the root of μ over $\mathbb{K}$. Then,

$$\mathbb{K} = \{c_0 + c_1\theta + \cdots + c_{n-1}\theta^{n-1} : c_0, \dots, c_{n-1} \in \mathbb{F}\}$$

Although this definition is quite useful, we will mostly rely on the polynomial quotient ring definition. Let us define the **prime field extension**.

Definition 0.4. Suppose p is prime and $m \geq 2$. Let $\mu \in \mathbb{F}_p[X]$ be an irreducible polynomial of degree m . Then, elements of $\mathbb{F}_{p^m}$ are polynomials in $\mathbb{F}_p^{(\leq m)}[X]$. In other words,

$$\mathbb{F}_{p^m} = \{c_0 + c_1X + \cdots + c_{m-1}X^{m-1} : c_0, \dots, c_{m-1} \in \mathbb{F}_p\},$$

where all operations are performed modulo $\mu(X)$.

Again, let us consider a few examples.

Example 0.8. Consider the $\mathbb{F}_{2^4}$. Then, there are 16 elements in this set:

$$\begin{aligned} &0, 1, X, X + 1, \\ &X^2, X^2 + 1, X^2 + X, X^2 + X + 1, \\ &X^3, X^3 + 1, X^3 + X, X^3 + X + 1, \\ &X^3 + X^2, X^3 + X^2 + 1, X^3 + X^2 + X, X^3 + X^2 + X + 1. \end{aligned}$$

One might choose the following reduction polynomial: $\mu(X) = X^4 + X + 1$ (of degree 4). Then, operations are performed in the following manner:

- Addition: $(X^3 + X^2 + 1) + (X^2 + X + 1) = X^3 + X$.
- Subtraction: $(X^3 + X^2 + 1) - (X^2 + X + 1) = X^3 + X$.
- Multiplication: $(X^3 + X^2 + 1) \cdot (X^2 + X + 1) = X^2 + 1$ since:

$$(X^3 + X^2 + 1) \cdot (X^2 + X + 1) = X^5 + X + 1 \pmod{X^4 + X + 1} = X^2 + 1$$
- Inversion: $(X^3 + X^2 + 1)^{-1} = X^2$ since $(X^3 + X^2 + 1) \cdot X^2 \pmod{X^4 + X + 1} = 1$.

Now, in the subsequent sections, we would need to extend $\mathbb{F}_p$ at least to $\mathbb{F}_{p^2}$. A convenient choice, similarly to the complex numbers, is to take $\mu(X) = X^2 + 1$. However, in contrast to $\mathbb{R}$, equation $X^2 = -1 \pmod{p}$ might have solutions over certain prime numbers p . Thus, we consider proposition below.

Proposition 0.5. Let p be an odd prime. Then $X^2 + 1$ is irreducible in $\mathbb{F}_p[X]$ if and only if $p \equiv 3 \pmod{4}$.

Corollary 0.6. $\mathbb{F}_{p^2} = \mathbb{F}_p[u]/(u^2 + 1)$ is a valid prime field extension for odd primes p satisfying $p \equiv 3 \pmod{4}$. In this case, extended elements are of the form $c_0 + c_1 u$ where $c_0, c_1 \in \mathbb{F}_p$ and $u^2 = -1$.

0.3 Multiplicative Group of a Finite Field

The non-zero elements of $\mathbb{F}_p$, denoted as $\mathbb{F}_p^\times$, form a multiplicative cyclic group. In other words, there exist elements $g \in \mathbb{F}_p^\times$, called *generators*, such that

$$\mathbb{F}_p^\times = \{g^k : 0 \leq k \leq p-2\}$$

The order of $x \in \mathbb{F}_p^\times$ is the smallest positive integer r such that $x^r = 1$. It is also not difficult to show that $r \mid (p-1)$.

Definition 0.7. $\omega \in \mathbb{F}$ is the *primitive root* in the finite field $\mathbb{F}$ if $\langle \omega \rangle = \mathbb{F}^\times$.

Example 0.9. $\omega = 3$ is the primitive root of $\mathbb{F}_7$. Indeed,

$$3^1 = 3, 3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5, 3^6 = 1.$$

So clearly $\langle \omega \rangle = 7$.

In STARKs (and in optimizing operations) for DFT (Discrete Fourier Transform) we would need the so-called n th primitive roots of unity.

Example 0.10. For those who studied complex numbers a bit (it is totally OK if you did not, so you might skip this example), recall an equation $\zeta^n = 1$ over $\mathbb{C}$. The solutions are $\zeta_k = \cos\left(\frac{2\pi k}{n}\right) + i \sin\left(\frac{2\pi k}{n}\right)$ for $k \in \{0, 1, \dots, n-1\}$, so one has exactly n solutions (in contrast to $x^n = 1$ over $\mathbb{R}$ where there are at most 2 solutions^a). For any solution ζ_k , it is true that $\zeta_k^n = 1$, but if one were to consider the subgroup generated by ζ_k (that is, $\{1, \zeta_k, \zeta_k^2, \dots\}$), then not necessarily $\langle \zeta_k \rangle$ would enumerate all the roots of unity $\{\zeta_j\}_{j=0}^{n-1}$. For that reason, we call ζ_k the n th primitive root of unity if $\langle \zeta_k \rangle$ enumerates all roots of unity. One can show that this is the case if and only if $\gcd(k, n) = 1$. This is always the case for $k = 1$, so commonly mathematicians use ζ_n to denote an expression $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} = e^{2\pi i/n}$.

^aThink why.

Yet, let us give the broader definition, including the finite fields case.

Definition 0.8. ω is the n th primitive root of unity if $\omega^n = 1$ and $\omega^k \neq 1$ for all $1 \leq k < n$.

Note that such ω exists if and only if $n \mid (p - 1)$.

0.4 Algebraic Closure

Consider the following interesting question: suppose we have a field $\mathbb{F}$. Is there an extension $\mathbb{K}/\mathbb{F}$ such that $\mathbb{K}$ contains all roots of any polynomial in $\mathbb{F}[X]$? The answer is yes, and such a field is called the **algebraic closure** of $\mathbb{F}$, although not always this algebraic closure has a nice form. But first, let us define what it means for field $\mathbb{F}$ to be algebraically closed.

Definition 0.9. A field $\mathbb{F}$ is called **algebraically closed** if every non-constant polynomial $p(x) \in \mathbb{F}[X]$ has a root in $\mathbb{F}$.

Example 0.11. $\mathbb{R}$ is not algebraically closed since $X^2 + 1$ has no roots in $\mathbb{R}$. However, $\mathbb{C}$ is algebraically closed, which follows from the fundamental theorem of algebra. Since $\mathbb{C}$ is a field extension of $\mathbb{R}$, it is also an algebraic closure of $\mathbb{R}$. This is commonly denoted as $\overline{\mathbb{R}} = \mathbb{C}$.

Definition 0.10. A field $\mathbb{K}$ is called an **algebraic closure** of $\mathbb{F}$ if $\mathbb{K}/\mathbb{F}$ is algebraically closed. This is denoted as $\overline{\mathbb{F}} = \mathbb{K}$.

Since we are doing cryptography and not mathematics, we are interested in the algebraic closure of $\mathbb{F}_p$. Well, I have two news for you (as always, one is good and one is bad). The good news is that any finite field $\mathbb{F}_{p^m}$ has an algebraic closure. The bad news is that it does not have a form $\mathbb{F}_{p^k}$ for $k > m$ and there are infinitely many elements in it (so in other words, the algebraic closure of a finite field is not finite). This is due to the following theorem.

Theorem 0.11. No finite field $\mathbb{F}$ is algebraically closed.

Proof. Suppose $f_1, f_2, \dots, f_n \in \mathbb{F}$ are all elements of $\mathbb{F}$. Consider the following polynomial:

$$p(x) = \prod_{i=1}^n (x - f_i) + 1 = (x - f_1)(x - f_2) \cdots (x - f_n) + 1.$$

Clearly, $p(x)$ is a non-constant polynomial and has no roots in $\mathbb{F}$, since for any $f \in \mathbb{F}$, one has $p(f) = 1$. ■

But what form does the $\overline{\mathbb{F}}_p$ have? Well, it is a union of all $\mathbb{F}_{p^k}$ for $k \geq 1$. This is formally written as:

$$\overline{\mathbb{F}}_p = \bigcup_{k \in \mathbb{N}} \mathbb{F}_{p^k}.$$

Remark. But this definition is super counter-intuitive! So here how we usually interpret it. Suppose I tell you that polynomial $q(x)$ has a root in $\overline{\mathbb{F}}_p$. What that means is that there exists some extension $\mathbb{F}_{p^m}$ such that for some $\alpha \in \mathbb{F}_{p^m}$, $q(\alpha) = 0$. We do not know how large this m is, but we know that it exists. For that reason, $\overline{\mathbb{F}}_p$ is defined as an infinite union of all possible field extensions.

0.5 Exercises

Exercise 1. Oleksandr decided to build $\mathbb{F}_{49}$ as $\mathbb{F}_7[i]/(i^2 + 1)$. Compute $(3 + i)(4 + i)$.

- a) $6 + i$. b) 6. c) $4 + i$. d) 4. e) $2 + 4i$.

Exercise 2. Oleksandr came up with yet another extension $\mathbb{F}_{p^2} = \mathbb{F}_p[i]/(i^2 + 2)$. He asked interns to calculate $2/i$. Based on five answers given below, help Oleksandr to find the correct one.

- a) 1. c) $(p - 3)i$. e) $p - 1$.
b) $p - 2$. d) $(p - 1)i$.

Exercise 3*. After endless tries, Oleksandr has finally found the perfect field extension: $\mathbb{F}_{p^2} := \mathbb{F}_p[v]/(v^2 + v + 1)$. However, Oleksandr became very frustrated since not for any p this would be a valid field extension. For which of the following values p such construction would **not** be a valid field extension? Use the fact that equation $\omega^3 = 1$ over $\mathbb{F}_p$ has non-trivial solutions (meaning, two others except for $\omega = 1$) if $p \equiv 1 \pmod{3}$. You can assume that listed numbers are primes.

- a) 8431. b) 9173. c) 9419. d) 6947.

Exercises 4-9. Tower of Extensions

You are given the passage explaining the topic of tower of extensions. The text has gaps that you need to fill in with the correct statement among the provided choices.

This question demonstrates the concept of the so-called **tower of extensions**. Suppose we want to build an extension field $\mathbb{F}_{p^4}$. Of course, we can find some irreducible polynomial $p(X)$ of degree 4 over $\mathbb{F}_p$ and build $\mathbb{F}_{p^4}$ as $\mathbb{F}_p[X]/(p(X))$. However, this method is very inconvenient since implementing the full 4-degree polynomial arithmetic is inconvenient. Moreover, if we were to implement arithmetic over, say, $\mathbb{F}_{p^{24}}$, that would make the matters worse. For this reason, we will build $\mathbb{F}_{p^4}$ as $\mathbb{F}_{p^2}[j]/(q(j))$ where $q(j)$ is an irreducible polynomial of degree 2 over $\mathbb{F}_{p^2}$, which itself is represented as $\mathbb{F}_p[i]/(r(i))$ for some suitable irreducible quadratic polynomial $r(i)$. This way, we can first implement $\mathbb{F}_{p^2}$, then $\mathbb{F}_{p^4}$, relying on the implementation of $\mathbb{F}_{p^2}$.

For illustration purposes, let us pick $p := 5$. As noted above, we want to build $\mathbb{F}_{5^2}$ first. A valid way to represent $\mathbb{F}_{5^2}$ would be to set $\mathbb{F}_{5^2} := \boxed{4}$. Given this representation, the zero of a linear polynomial $f(x) = ix - (i + 3)$, defined over $\mathbb{F}_{5^2}$, is $\boxed{5}$.

Now, assume that we represent $\mathbb{F}_{5^4}$ as $\mathbb{F}_{5^2}[j]/(j^2 - \xi)$ for $\xi = i + 1$. Given such representation, the value of j^4 is $\boxed{6}$. Finally, given $c_0 + c_1j \in \mathbb{F}_{5^4}$ we call $c_0 \in \mathbb{F}_{5^2}$ a **real part**, while $c_1 \in \mathbb{F}_{5^2}$ an **imaginary part**. For example, the imaginary part of number $j^3 + 2i^2\xi$ is $\boxed{7}$, while the real part of $(a_0 + a_1j)b_1j$ is $\boxed{8}$. Similarly to complex numbers, it motivates us to define the number's **conjugate**: for $z = c_0 + c_1j$, define the conjugate as $\bar{z} := c_0 - c_1j$. The expression $z\bar{z}$ is then $\boxed{9}$.

Exercise 4.

- $\mathbb{F}_5[i]/(i^2 + 1)$
- $\mathbb{F}_5[i]/(i^2 + 2)$
- $\mathbb{F}_5[i]/(i^2 + 4)$
- $\mathbb{F}_5[i]/(i^2 + 2i + 1)$
- $\mathbb{F}_5[i]/(i^2 + 4i + 4)$

Exercise 7.

- equal to zero.
- equal to one.
- equal to the real part.
- $2(1 + i)$
- -4

Exercise 5.

- $1 + i$
- $1 + 2i$
- $1 + 4i$
- $2 + 3i$
- $3 + i$

Exercise 8.

- a_1b_1
- $a_1b_1\xi$
- a_0b_1
- $a_0b_1\xi$
- a_0a_1

Exercise 6.

- $4 + 2i$
- $4i$
- 1
- $1 + 2i$
- $2 + 4i$

Exercise 9.

- $c_0^2 + c_1^2$
- $c_0^2 - c_1^2\xi$
- $c_0^2 + c_1^2\xi^2$
- $(c_0^2 + c_1^2\xi)j$
- $(c_0^2 - c_1^2)j$